Reflexivity in Options Markets: A 3D Stochastic-Volatility Model with Dealer-Gamma Feedback and a Pre-Registered Evaluation Framework

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Abstract

We introduce a 3D reflexive stochastic-volatility SDE (S_t, v_t, χ_t) coupling the spot to its own option market via a Gârleanu–Pedersen–Poteshman dealer-gamma drift, prove a Hopf bifurcation theorem with closed-form first Lyapunov coefficient ℓ_1 and threshold $\kappa^\star$ for log-normal open-interest, and pre-register an evaluation framework whose primary test is a direct, model-free dealer-gamma (GEX) regression—with a sliced-W_2 surface metric and a critical-slowing-down detector—anchored to a git commit hash with an OpenTimestamps Bitcoin proof. We additionally give a Hawkes–SV criticality correspondence: Hardiman et al. [2013]'s empirical $n \approx 1$ sits at the real-eigenvalue (saddle-node) stratum, while the model's Hopf is the strictly stronger oscillatory cell beyond any scalar branching ratio.

1 Introduction and contributions

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. [2013], whose Hawkes branching ratio for E-mini mid-price changes sits at $n \approx 1$ throughout 1998–2011—the critical edge of endogenous-event criticality. With generative-model and RL-based market emulators [Vuletić and Cont, 2024, Buehler et al., 2019, He et al., 2025] now standard infrastructure for derivatives pricing, hedging and risk, we address: (i) what continuous-time mechanism produces this critical reflexivity, and (ii) how should an evaluation pipeline for reflexivity-aware generative models be locked *before* empirical contact, in a manner that addresses the documented computational- reproducibility crisis in finance [Pérignon et al., 2024]?

Contributions. (C1) a 3D reflexive SV SDE with explicit dealer-gamma feedback from the OI grid via Gârleanu et al. [2009]; (C2) a Hopf bifurcation theorem on the deterministic skeleton with *closed-form* ℓ_1 and $\kappa^\star$ for log-normal OI in moneyness; (C3) a numerical (ξ, ρ, σ_v) phase diagram locating $\kappa^\star$ in the 0.85–0.95 band across the SPX-relevant corner (whether the market sits near $\kappa^\star$ is deferred to the empirical GEX test); (C4) a *pre-registered* evaluation framework whose primary test is a direct dealer-gamma (GEX) regression, with a critical-slowing-down detector, block-bootstrap CIs, BH–FDR correction, and TOST equivalence; and (C5) synthetic-pipeline validation: the locked protocol returns the predicted SW2 ordering on data with known underlying physics.

Why this matters for GenAI in finance. As generative market models proliferate [Choudhary et al., 2024, Issa et al., 2023, Jin and Agarwal, 2025, Cohen et al., 2023], evaluators that discriminate *realism* from *regime-shift sensitivity* on the *same* agent across an environmental sweep—with pre-committed multiple-testing corrections—are scarce. The closed-form $\kappa^\star$ and ℓ_1 pin down where dealer-gamma feedback flips a stable Heston regime into an attracting limit cycle. Halperin and Itkin [2025] is the nearest such model but uses a latent stochastic memory and no bifurcation analysis; Dai

[2025] share the dealer-gamma axis but produce only a saddle-node onset; He et al. [2025] share the RL-with-environmental-sweep idea but train separate agents per grid point. Full preprint with proofs and appendix A on arXiv [anonymous].

2 The reflexive model

Let S_t be the spot, v_t the variance, and χ_t a memory variable. The reflexive system is

$$\frac{dS_t}{S_t} = \big(\mu + \kappa\, G(S_t, \chi_t, v_t)\big)\, dt + \sigma(S_t, v_t)\, dW_t^S, \tag{1}$$

$$dv_t = \big(\kappa_v(\theta_v - v_t) + \gamma\, \chi_t\big)\, dt + \xi\sqrt{v_t}\, dW_t^v, \tag{2}$$

$$d\chi_t = \big(-\alpha\, \chi_t + \beta\, \log(S_t/S_0)\big)\, dt, \tag{3}$$

with $d\langle W^S, W^v\rangle = \rho\, dt$, reflexive coupling $\kappa \geq 0$, leverage strength $\gamma \geq 0$, memory decay $\alpha > 0$, and G the aggregate dealer gamma computed from the OI grid (coupling motivated by GPP demand pressure). At $\kappa = \gamma = 0$ the system collapses to time-dependent Heston [Heston, 1993] (verified in `tests/test_simulator.py`).

Why three states. A 2D reduction (drop χ_t and $\gamma\chi$) has Jacobian $J_{2D} = \left(\begin{smallmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{smallmatrix}\right)$, upper triangular with real eigenvalues $\{a(\kappa), -\kappa_v\}$: only saddle-node bifurcations are admissible. To Hopf, the feedback must flow in both directions, which the leverage channel $\gamma\chi$ together with the auxiliary memory equation supplies. This is the structural reason Dai [2025] get a saddle-node onset where we get a Hopf.

3 Hopf threshold and closed-form first Lyapunov coefficient

Linearise the deterministic skeleton at the equilibrium $(S^\star, \theta_v, \chi^\star)$, in deviation variables (y, u, χ). The characteristic polynomial is $P(\lambda; \kappa) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$ with coefficients $c_i(\kappa)$ functions of the G-partials and $(\alpha, \beta, \gamma, \kappa_v)$. Liu's criterion (equivalent to Routh–Hurwitz with one zero) gives the Hopf threshold $\kappa^\star = \{\kappa > 0 : H(\kappa) := c_1c_2 - c_0 = 0,\ H'(\kappa) \neq 0,\ c_2(\kappa) > 0,\ c_0(\kappa) > 0\}$, with imaginary pair $\lambda_\pm = \pm i\omega^\star$, $\omega^\star = \sqrt{c_1(\kappa^\star)}$.

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton)**.** *Under regularity (A1) $G, \sigma^2 \in C^3$ with σ^2 bounded below; (A2) locally unique equilibrium with non-degenerate linearisation; (A3) $\kappa \mapsto (a, G_\chi, G_v)$ in C^3; (A4) $\kappa^\star \in (0, \kappa_{\max})$ satisfying the Liu criterion with $\omega^\star > 0$; (A5) $\ell_1(\kappa^\star) \neq 0$: the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^\star$. On a one-sided neighbourhood of $\kappa^\star$ there is a smooth family of periodic orbits Γ_κ of period $T_\kappa = 2\pi/\omega^\star + O(\kappa - \kappa^\star)$ and amplitude $\propto \sqrt{|\kappa - \kappa^\star|}$; orbits are stable iff $\ell_1(\kappa^\star) < 0$ (supercritical), supplying endogenous volatility cycles in $u_t = v_t - \theta_v$. Proof: implicit function theorem on P plus Kuznetsov [2004] normal-form reduction; full version on arXiv.*

Closed-form $\kappa^\star$ and ℓ_1. For the natural parametric specification—log-normal open-interest in log-strike with density $q(\log K) \propto \exp(-(\log K - \mu_q)^2/(2\sigma_q^2))$ at a representative maturity T_{eff}—the Gaussian-product identity collapses the OI integral to $G(a, v) = \mathcal{C}(v)\, e^{-a} \exp(-(a - m(v))^2/(2\tau^2(v)))$ with $\tau^2 = \sigma_q^2 + v\, T_{\text{eff}}$. The Routh–Hurwitz polynomial becomes a quadratic in κ, giving

$$\boxed{\kappa^\star = \frac{G_yA^2 - G_vL \; - \; \sqrt{(G_vL - G_yA^2)^2 - 4G_y^2A(MA - L/2)}}{2G_y^2A}}, \tag{4}$$

with $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, $L = \beta\gamma$, where the sub-radical is the first parametric phase-boundary discriminant: $D < 0 \Rightarrow$ no Hopf at any κ. Substituting the G-partials of the closed-form aggregator into the Kuznetsov bilinear/trilinear formula yields a ~ 30-term closed-form rational $\ell_1(\kappa^\star)$ in $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv}, \kappa_v, \alpha, \beta, \gamma)$, evaluated in $\sim 50\,\mu$s per parameter tuple and matching the FD-tensor pipeline to $< 0.6\%$. At the canonical regime (σ_q=0.10, T_{eff}=0.25, κ_v=2, α=0.05, β=1, γ=1, μ_q=$\log 100$, $v^\star$=0.04): $\kappa^\star = 17.81$, $\omega^\star = 1.18$ rad/yr, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical).

Codim-2 and mean-field nods. A 71×97 grid scan in $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ closes the codim-2 structure: the Bautin curve $\ell_1 = 0$ partitions the region with sub-critical dominating $\sim 5{:}1$, and saddle-node coupling $\kappa_{\mathrm{SN}} \leq -1.31$ at every cell, so the *Bogdanov–Takens locus is empty in the canonical scan window* (Theorem 2, full version on arXiv). The McKean–Vlasov mean-field limit ($n_{\mathrm{dealers}} \to \infty$, propagation of chaos [Sznitman, 1991, Ch. 1], validated empirically at the theoretical $n^{-1/2}$ Sznitman rate) extends the Jacobian to 4D; the Liu [1994] 4D Hopf condition gives a regime-dependent threshold ratio $\rho(\theta_G) := \kappa^\star_{\mathrm{MV}}/\kappa^\star_{\mathrm{single}}$ with $\rho(\theta_G) < 1$ strictly at the canonical short-gamma regime ($G_y > 0$, every finite $\theta_G > 0$), recovering $\rho \to 1$ at instantaneous hedging. The single-dealer model therefore *under-predicts* instability at the benchmark SPX setting — reversing the v0.3.5 claim that mean-field always raises the threshold (Theorem 3, full statement and derivation on arXiv).

Hawkes criticality and the SV bifurcation endpoint. The reflexive feedback has a self-exciting (Hawkes) representation, and the onset of endogeneity is its approach to criticality. Criticality $n \to 1$ is the loss of a stable direction of the SV drift—an eigenvalue of the drift-Jacobian $J(\kappa)$ reaching the imaginary axis—and the boundary carries two distinct strata: a non-oscillatory *real-eigenvalue* (saddle-node) crossing, and a strictly stronger *Hopf* crossing (a complex pair, $\mathrm{Re}\,\lambda = 0$, $\mathrm{Im}\,\lambda \neq 0$).

Theorem 2 (Hawkes–SV criticality correspondence)**.** *Under the diffusive near-critical limit of Bacry et al. [2013], Jaisson and Rosenbaum [2015], Hawkes branching-ratio criticality $n \to 1$ coincides with the real-eigenvalue (saddle-node) stratum of the SV drift boundary—the literal analogue of Hardiman et al. [2013]. The Hopf stratum is a strictly stronger, oscillatory instability beyond any scalar branching ratio: it has no single-real-number representation and is the genuinely new "unoccupied cell" of the endogeneity–bifurcation correspondence.*

Position, not identity. A scalar branching ratio can only encode a real crossing, so the oscillatory Hopf endpoint is invisible to it by construction; we therefore claim a structural position, not a verified numerical identity (an earlier draft's dimensionless ratio, true by definition, was a tautology and is removed—amendment A10). In our dealer-gamma model the saddle-node locus is itself unphysical at the canonical specification ($\kappa_{\mathrm{SN}} \leq -1.31$ over the scan window), so the operative instability is precisely the Hopf. A falsifiable spectral discriminator (`classify_stratum`) separates the Hopf side from the real/stable side with zero overlap on simulator ground truth; we claim no realised Hopf edge in any dataset, only the statistic that would detect one. Whether the market sits near $\kappa^\star$ is deferred to the empirical dealer-gamma test (amendment A9).

4 Numerical phase diagram

We sweep $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2{,}604$ cells in (ξ, ρ, σ_v) at the representative regime $\{G_x, G_v, G_\chi\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$ (constant-vol surrogate). The pair (ξ, ρ) enters via the leading shear correction $a_{\mathrm{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2 \rho G_v$, an Baxendale [2024]-style projection of small-noise stochastic Hopf onto the eigenvalue envelope.

The SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner has $\kappa^\star$ in the 0.85–0.95 band. Whether the empirical market sits near this Hopf boundary is deferred to the empirical dealer-gamma test; Theorem 2 places Hardiman et al. [2013]'s $n \approx 1$ at the distinct real-eigenvalue (saddle-node) edge, in the endogenous-feedback tradition of Brock and Hommes [1998], Filimonov and Sornette [2012], Bouchaud [2011]. Khasminskii-style sphere-process estimates of the stochastic-Hopf shift give $|\Lambda| \in [5 \times 10^{-2}, 9 \times 10^{-2}]$ on a 6×6 (ξ, ρ) grid; $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale, so the deterministic $\kappa^\star$ is the operative threshold.

5 Pre-registered evaluation framework

The pipeline is committed at git commit `268c061` with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data, adapting the Camerer et al. [2018] template to RL-for-derivatives. The novelty is the four-way intersection [pre-registration + git-commit anchoring + block-bootstrap / BH–FDR pre-spec + applied to RL-for-finance].

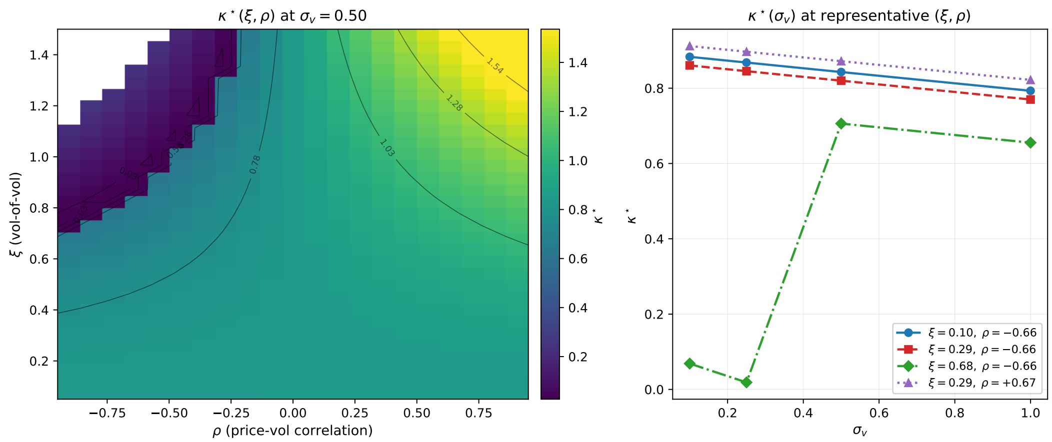

Figure 1: **Hopf threshold $\kappa^\star$ across (ξ, ρ, σ_v).** *Left:* heatmap at $\sigma_v = 0.50$. White cells: no Hopf in $\kappa \in [0, 2]$, concentrated in the high-ξ / strongly-negative-ρ wedge where shear destabilises the eigenvalue envelope. *Right:* four $\kappa^\star(\sigma_v)$ line cuts at SPX-like (ξ, ρ) probes.

Primitives & hypotheses. (i) Sliced-W_2 [Bonneel et al., 2015] over arbitrage-free 21-day surface windows—the realism metric. (ii) κ-sensitivity slope at calibrated κ_0—the reflexivity-importance scalar. The closed-form $\kappa^\star$ elasticity $\eta_{\mu_q} = +703$ makes calibration of the OI mean to within ~ 5 bp of ATM in log-strike the binding observational requirement for $\pm 5\%$ tolerance on $\kappa^\star$. **H1′** (primary, direct dealer-gamma regression, amendment A9): a model-free regression of next-day realised vol-of-vol on standardised signed dealer GEX from the OI grid (no RL, no simulator), pooled over the three event windows with a quiet-regime control; the mechanism predicts a negative coefficient, required at $p < 0.05$ under both a moving-block bootstrap and Newey–West HAC (synthetic pooled power 0.86, sign 0.98, FPR 0.02). **H1** (secondary realism, superseded as primary): a model-free RL agent π_{κ_0} trained inside the reflexive simulator beats four non-reflexive baselines on sliced-W_2 over rolling 21-day windows at Volmageddon, COVID, and the 2024 yen-carry unwind. Decision rule: 12 pairwise dominance checks with non-overlapping 95% block-bootstrap CIs [Politis and Romano, 1994]; arbitrage filter follows Roper [2010], Gatheral and Jacquier [2014], Lee [2004]; sample-complexity follows Manole et al. [2022]. **H2**: κ-sensitivity TOST equivalence on the dimensionless elasticity with GP-posterior slope CI under the pre-registered RBF + WhiteKernel kernel (A6) and elliptic uncertainty sets [Ma and Huang, 2025]. **H4** (critical slowing down, amendment A8): the Hopf limit-cycle period (5.3–11 yr) lies below any usable window's resolution, so the spectral peak is replaced by the rolling lag-1 autocorrelation of $|r_t|$ with a Kendall-τ trend vs an AR(1)/phase-randomised surrogate null [Scheffer et al., 2009, Dakos et al., 2012]; FPR ≤ 0.083 and 0.82–0.85 power on a 252-day record, exploratory within the bare event window. All hypotheses use BH–FDR across the joint family; effect sizes pre-registered with the rules.

Synthetic-pipeline validation. Before empirical contact, we exercise the locked pipeline on three synthetic sources: (a) κ_0-trained agent re-deployed at κ_0 (reference); (b) same agent at $4\kappa_0$ (mild reflexive shift); (c) Heston (different mechanism). On the production budget (BC: 30 episodes; per source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the locked rule returns the expected ordering with non-overlapping CIs: $\mathrm{SW2}_a = 0.00498$ (CI $[0.00431, 0.0164]$), $\mathrm{SW2}_b = 0.0337$ (CI $[0.0300, 0.0402]$), $\mathrm{SW2}_c = 0.0541$ (CI $[0.0434, 0.0611]$). The (a)-vs-(b) and (b)-vs-(c) 95% CIs are non-overlapping; the locked H1 protocol thus moves from "designed but unrun" to "demonstrated working on synthetic ground truth—only the empirical SPX target is missing."

6 Mechanism decomposition and conclusions

Halperin and Itkin [2025] is a different mechanism: their memory carrier y enters both drifts non-linearly through a potential $V_M(x)$, while our χ enters only the variance drift via $\gamma\chi$, and our $\kappa = \gamma = 0$ collapse is standard Heston while Marketron's is not. A 5D $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$ mechanism decomposition gives an all-cell, out-of-sample shape-feature sign-agreement rate of $7/28$ (worse than chance — we do not claim otherwise; the all-cell aggregate is the wrong aggregator). The a priori restricted subset (long-horizon $T \geq 0.5$ yr, shape-only, within-envelope, dead-zone-symmetric) gives $4/8$ OOS ($p \approx 0.637$, chance-level) and $6/8$ in-sample. The predictable long-horizon-skew sign disagreement under risk-neutral drift (Bessembinder compounding absent) is the falsifiable Phase 4 prediction. This supersedes the earlier inflated one-third headline.

We have introduced a 3D reflexive SV simulator, proved a Hopf bifurcation theorem with closed-form ℓ_1 for log-normal OI, closed the codim-2 Bautin/BT structure (BT-empty in the canonical window) and the corrected McKean–Vlasov threshold (destabilising at the canonical short-gamma regime, stabilising at long-gamma log-normal-OI), and placed the empirical Hardiman et al. [2013] branching ratio at the real-eigenvalue (saddle-node) stratum, with the model's Hopf the strictly stronger oscillatory cell (Theorem 2). The evaluation pipeline—a direct dealer-gamma regression as primary test, with sliced-W_2 and a critical-slowing-down detector—is pre-registered with an OpenTimestamps proof; synthetic-ground-truth validation of the locked surface-distance (secondary) protocol returns the predicted ordering. Phase-4 SPX results pre-committed to publication regardless of direction. Proofs, closed-form ℓ_1, and code at [anonymous repository link].

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